

CONTINUATION — Helix Proxy: VPN-Aware Dynamic Routing Extension

Revision: 5 **Last modified:** 2026-07-01T11:25:00Z **Status:** Active — anti-bluff existing-suite sweep COMPLETE (BUGFIX-0014...0018 landed + guarded). Let's Encrypt workstream (task #59): **Phase 0/1/2/3 DONE + committed** — a REAL hermetic DNS-01 certificate is issued by local Pebble and cert-analyzer-verified (re-runnable deploy/letsencrypt/phase3_hermetic_issue.sh, live-PASS Challenge, RED+GREEN-proven guard) @ b2afa7d. **Task #56 observability live-/metrics-scrape DONE + committed** (isolated trio, real Prometheus exposition proven) @ 401234e. **LE Phase 5 (renewal/rotation) root-diagnosed but PENDING** — Pebble $\geq 2.8.0$ /set-renewal-info works, but certmagic v0.21.3's persistent 6h ARI cache blocks a fast deterministic renewal; clean fix = Caddy $\geq 2.11.0$ + libdns v1.0.0 provider port (feasibility research in flight, docs/research/caddy_2110_ari_refetch_20260701/). LE Phase 4 (staging token) + Phase 6 (prod cutover) + the LIVE dynamic-VPN P10 proof remain operator-gated. HEAD cc58a0c. **Branch:** feature/vpn-aware-dynamic-routing **Spec:** docs/superpowers/specs/2026-06-30-vpn-aware-proxy-extension-design.md (Rev 4) **Plan:** docs/superpowers/plans/2026-06-30-vpn-aware-proxy-extension-plan.md (Rev 1) **Authority:** Inherits the Helix Constitution submodule (constitution/Constitution.md) per §11.4.35.

§12.10 live-state resume file. Read this first, then `git fetch --all --prune` and re-read `git log --oneline main..HEAD`. Any agent must be able to resume exactly where the last session left off from this single file.

1. Current PHASE

Construction phase — landed: P0-P7 (control-plane + config-plane + control-API). 28 commits on feature/vpn-aware-dynamic-routing ahead of main. The Go control-plane (stores, health-publisher, acl-helper, config-compiler, P5b breaker/failover, P6 control-API/SSE/metrics/PAC/mTLS) builds clean (4 binaries: acl-helper, api, compiler, healthd; `go build/go vet/gofmt -l` all clean) and is proven at the unit / integration / config-parse layer. The

dynamic compose profile + Containerfiles + orchestrator wiring are authored (P10-prep) but **never booted** — the live dynamic-VPN data-plane proof is the entire job of **P10**.

MAJOR live finding this session (BUGFIX-0002): the existing proxy genuinely did NOT serve under rootless Podman — squid crash-looped because the host-created ./logs bind-mount was mode 0755 and the container's remapped non-root proxy user could not write access.log (FATAL). Fixed (chmod 777 the log dir, mirroring the existing cache-dir remedy). After the fix the **3 existing features are PROVEN-WORKING-LIVE:** HTTP forward proxy (200 + Via: proxy-squid), Dante SOCKS5 (200), squid caching (TCP_MEM_HIT, no origin contact). This directly answers the operator's "most features don't work / can't be used" concern at the existing-proxy layer.

BUGFIX-0003: run-tests.sh's test_result leaked a non-zero status on a no-message FAIL, aborting the suite after ~10 of 41 tests under set -e. Fixed (return 0); the suite now runs to completion and honestly reports 7 pre-existing FAILs (the §11.4.1 false-FAILs P8 is fixing now).

P8 anti-bluff existing-test sweep — COMMITTED as four bugfix lanes: BUGFIX-0004 (cd11494, run-tests.sh §11.4.3 topology-aware ports + 3-state SKIP + §11.4.161 podman), BUGFIX-0005 (4394643, final-verify.sh + verify-proxy.sh false-VPN-routing §15 + set -e abort), BUGFIX-0006 (2bc03de, comprehensive-test.sh revived from a 100%-dead (()) abort + real data-plane evidence for B2/B3/B8 + B1 honest VPN SKIP). **P6 follow-ups — COMMITTED** (8d95f8a, WARNING-3 real bidirectional metric-name drift guard, independently §1.1-mutation-proven; WARNING-4 concurrency consistency test + documented cross-step audit gap → #52; WARNING-5 plaintext-metrics-listener design note → #53).

MAJOR finding from the BUGFIX-0006 revival (#50, in flight): reviving the dead comprehensive-test.sh immediately surfaced a **real regressed feature** — the documented 368-line cache management CLI (./cache stats|clear|invalidate| trim, README/USER_GUIDE/CACHE/TROUBLESHOOTING) is **gone from HEAD**. §11.4.124 git-history investigation (FACT): commit 6ec58ef accidentally deleted it — the tracked cache FILE collided with the runtime cache/ DATA dir at the same path (CACHE_DIR=\$PROJECT_ROOT/cache, .gitignore:16 cache/). A background subagent is restoring it as non-colliding ./cachectl (§11.4.101 safe/reversible), live-testing it on the running proxy, adding a §11.4.135 guard + §11.4.18 companion, doc-syncing the 4 docs, and flipping the 4 comprehensive-test.sh cache-CLI SKIPS → real PASS — conductor reviews + commits on return (§11.4.142).

Remaining: #50 (cachectl restore, in flight) → **P10** (live dynamic boot — the usability proof; fail-closed half is data-plane-only + autonomous via a self-generated throwaway helixproxy_pg_password secret, real-VPN-egress half operator-gated on gluetun WireGuard creds) → **P12** (whole-branch review + full retest + merge no-force + prefixed tag). Follow-ups: #52 (store-tx audit atomicity), #53 (plaintext metrics listener, operator-gated).

2. The 28 landed commits (git log --oneline main..HEAD, newest first; this-session wave = newest 11)

#	Commit	Phase	Summary
28	8d95f8a	P6	real bidirectional metric-name drift guard (§1.1-mutation-proven) + concurrency consistency test; WARNING-3/4/5
27	2bc03de	BUGFIX-0006	revive + de-bluff comprehensive-test.sh ((()) abort = 100% dead) + real B2/B3/B8 evidence; surfaced regression #50
26	4394643	BUGFIX-0005	final-verify.sh + verify-proxy.sh no longer green a NO-VPN config (false-VPN-routing §15) + set -e abort
25	cd11494	BUGFIX-0004	run-tests.sh no longer FAILs a healthy proxy — §11.4.3 topology-aware ports + 3-state SKIP
24	6a8f886	P11	refresh CONTINUATION to live state (Rev 2) — 23 commits, P5b/ P6/BUGFIX-0002/0003 landed, P8 in flight
23	61b4215	chore	gofmt-format 6 pre-existing files (formatters-clean mandate; semantics-null verified)
22	62b22fe	P6	control-API server (REST/SSE/ metrics/PAC, fail-closed mTLS) + coherent operator-wiring contract
21	1045dfd	BUGFIX-0003	test_result must return 0 — suite no longer aborts mid-run under set -e
20	c6f2935	P9	§11.4.18 operator-guide companions for the 16 tests/ dynamic scripts

#	Commit	Phase	Summary
19	b5573a9	BUGFIX-0002	squid log-dir writable under rootless Podman (proxy crash-loop) — existing features now serve live
18	0aca034	P9	anti-bluff dynamic-routing test/analyzer harness (tests/dynamic)
17	e6e93ec	P10-prep	dynamic compose profile + control-plane/squid Containerfiles + orchestrator wiring
16	6bdeef9	P5b	circuit-breaker + tier-failover (internal/breaker, gobreaker/v2)
15	7d0d128	P11	CONTINUATION (§12.10) + spec §9 reconcile + §11.4.65 HTML/PDF export backfill
14	1833c8f	P7.3	per-user Squid auth + rootless Podman-secret loader + kill-switch design (no secrets)
13	603e039	P5a	acl-helper — Squid external_acl OK/ERR from Redis, fail-closed (stdlib)
12	e6e336f	P7.1	per-tunnel DoH/DoT (dnsproxy) config plan + DNS-leak test design
11	04526dd	P4	config-compiler — render Squid/Dante/PAC from PG + seed route keys (parse-verified)
10	833fb9e	P7.2	Prometheus scrape + Grafana dashboard config plan (promtool-validated)
9	11106a4	P3	vpn-health-publisher (cmd/healthd + internal/vpn) — data-plane health, fail-closed, TDD
8	b66d172	P4	Squid 6.13 + Dante dynamic-mode templates (additive, parse-verified) + spec reconcile
7	fbfe9ed	P1	docs(spec): mark §20 gaps G1-G4 RESOLVED with spike decisions
6	e19e0ed	P2	store (pgx) + redis (go-redis) clients — fail-closed, TDD, real PG/Redis
5	6409cb9	P1	docs(research): resolve spec §20 gaps G1-G4 with captured-evidence spikes
4	6802798	P1/E	

#	Commit	Phase	Summary
			docs(audit): §11.4.138 forensic bluff-audit of 4 existing test scripts (8 bluffs)
3	9ac1b4a	P0	docs(dynamic-routing): DYNAMIC_ROUTING.md + 2 mermaid diagrams
2	6251007	P0	chore(submodules): incorporate containers, helix_qa, challenges, docs_chain (SSH, no-force)
1	5f917a7	P0	P0 scaffold — data model, evidence harness, Go skeleton, governance carriers

3. PROVEN-NOW vs OWED-TO-P10 (honest §11.4.6)

PROVEN-NOW (control-plane / config-plane / spike facts — captured)

- **Existing proxy serves LIVE (BUGFIX-0002)** — after the rootless-Podman log-dir fix, the booted --no-vpn stack proves all 3 existing features: HTTP forward proxy 200 + Via: 1.1 proxy-squid, Dante SOCKS5 200, squid cache TCP_MEM_HIT (no origin contact). Guard: tests/regression/log_dir_writable_test.sh (§11.4.115 polarity, §1.1 mutation byte-identical md5 0128a96b6d467c2da5b7cef8a808e563). Evidence: qa-results/regression/bugfix38/.
- **P5b breaker/failover** (internal/breaker, gobreaker/v2) — per-target circuit breaker + tunnel tier-failover, TDD.
- **P6 control-API** (cmd/api + internal/api + internal/pac) — REST CRUD + SSE + Prometheus /metrics + PAC, **fail-closed mTLS** (RequireAndVerifyClientCert), coherent operator-wiring contract (CONTROL_API_TLS_CERT/_KEY/_TLS_CLIENT_CA, :58080); builds + vets clean, §1.1 mutation md5 67125c7a1ab9b00c98fb164f765b04af.
- **Spec §20 gaps G1-G4 resolved** with transient-spike captured evidence (docs/research/mvp/findings/F_spikes_G1-G4.md, run-id qa-results/spikes/20260630_205029_g1g4/): G2 ubuntu/squid:latest = Squid **6.13** (not v8), §8 directive set squid -k parse exit 0; G4 gluetun **v3.40 (=v3.40.4)** control-API :8000 answers 200, issue #3060 confirmed; G1 kernel-WG interface **creatable rootless** with --cap-add NET_ADMIN; G3 Dante **SIGHUP preserves an active SOCKS session** (20/20 chunks, curl exit 0, /proc/net/tcp ESTABLISHED proof).
- **P2 stores** (pgx + go-redis) — fail-closed, TDD, exercised against **real PG / Redis**.

- **P3 vpn-health-publisher** (cmd/healthd + internal/vpn) — data-plane health poll → Redis state, fail-closed, TDD.
- **P4 config-compiler + templates** — Squid 6.13 (%>ha{Host}) + Dante (concatenation, no include) render from PG; **squid -k parse exit 0**; PAC + route-key seeding parse-verified.
- **P5a acl-helper** — Squid external_acl OK/ERR from Redis, **fail-closed**, stdlib-only.
- **P7.2 observability** config plan — **promtool-validated** Prometheus scrape + Grafana dashboard.
- **P7.1 DNS / P7.3 security** — config plans only (DoH/DoT per-tunnel; per-user auth + Podman-secret loader + in-netns kill-switch). Design + parse layer.
- **Existing-test bluff audit** (Stream E) — 8 bluffs across 4 scripts catalogued (§11.4.138), guards owed to P8.

OWED-TO-P10 (the LIVE dynamic-stack data-plane proof — NOT yet captured)

The dynamic compose profile (postgres + redis + control-plane + gluetun(s) + squid+helper + dante) has **never been booted**; docs/DYNAMIC_ROUTING.md is explicitly DESIGN-only. The following remain **unproven live** and are the usability proof:

- **vpn_real_egress** — egress IP via proxy
== tunnel exit && != host IP + **wg transfer Δ** (200 OK is not routing).
- **graceful_503** — tunnel down → branded 503 with **Squid PID unchanged** → up → 200, live.
- **no_leak / killswitch** — drop tunnel → **zero** target packets on the real uplink (tcpdump) + DNS only via the intended resolver.
- per-user **407 auth challenge** live; **secret injection leak-free** at runtime.
- **G1 residual** — full rootless kernel-WG *operation* (handshake + routing + throughput), only interface *creation* was spiked (§20 G1).
- **G3 residual / P9** — concurrent / repeated SIGHUP + **route-change-mid-session** SOCKS path behaviour (§20 G3).
- circuit-breaker open → failover to next up tier (P5b not yet landed).

4. Remaining phases

Phase	Scope	State
P5b	per-target circuit breaker + tunnel tier-failover (sony/gobreaker/v2)	✅ landed 6bdeef9
P6	control-API + SSE + metrics + PAC + fail-closed mTLS	✅ landed 62b22fe (admin-UI templ/htmx + §11.4.170 host-rendered)

Phase	Scope	State
		pixel proof = P6.2, deferred)
P8	fix existing-test bluffs → §11.4.3 topology dispatch / honest SKIP / §11.4.161 + §11.4.135 guards	in flight (2 subagents: run-tests.sh + comprehensive-test.sh)
P9	full test matrix + Challenges + HelixQA (all §11.4.169 types; G3 route-change-mid-session live test)	harness landed 0aca034; live execution coupled to P10
P10	live dynamic-mode boot + captured data-plane evidence = the usability proof	critical unblocker ; fail-closed half doable now, real-egress half operator-gated (gluetun WG creds)
P11	docs sync + HTML/PDF (+DOCX where mandated) exports (this CONTINUATION + .remember are part of it)	ongoing
P12	whole-branch review (iterate-to-GO) + full retest + merge to main no-force + prefixed release tag	last

5. Binding constraints (non-negotiable)

- **Anti-bluff §11.4** — every PASS carries positive captured **data-plane** evidence; control-plane/config-parse green is necessary, never sufficient; the end-user-usability bar is met only at P10.
- **No force-push §11.4.113** — merge onto latest main, fast-forward only; force-push is forbidden with no exception.
- **Rootless Podman §11.4.161** — all containers rootless; no Docker-rootful, no sudo, no root escalation; orchestrate via the containers submodule (§11.4.76), build on the remote host (§11.4.173).
- **Secrets-as-names-only §11.4.10** — VPN creds / proxy-auth / mTLS keys via Podman secrets / file refs; **never** plaintext in git; .env.example documents refs only.
- **Operator-safe §11.4.174** — do **NOT** touch the operator's pre-existing resources: the host wg0-mullvad (UP kernel-WG) interface and any lava-* containers (e.g. lava-postgres-thinker) are off-limits; verify process/resource ownership before acting; block-don't-break on shared-host contention.

- **Host safety §12** — $\leq 60\%$ memory (§12.6); no host power-state commands (CONST-033); pull images sequentially; --rm diagnostics; df first.

6. Resume now (next actionable)

1. `git fetch --all --prune` on feature/vpn-aware-dynamic-routing; confirm HEAD cc58a0c (== main == all 3 remotes github/origin/upstream; integrate any newer foreign commit per §11.4.71, no force §11.4.113). The single canonical moment-valid resume file is `.remember/remember.md` (§11.4.131) — read it first.
2. **Anti-bluff sweep + LE Phase 0-3 + task #56 — ALL DONE + committed.** LE hermetic DNS-01 issuance is PROVEN (real cert from local Pebble, cert-analyzer-verified, re-runnable `deploy/letsencrypt/phase3_hermetic_issue.sh` @ b2afa7d; live-PASS Challenge `challenges/scripts/le_phase3_issuance_challenge.sh`; standing guard tests/`letsencrypt/phase3_issuance_guard.sh` RED+GREEN, wired in `run-tests.sh`). Task #56 observability live-/metrics-scrape PROVEN (isolated trio `deploy/observability/compose.metrics.yml`, real Prometheus exposition @ 401234e). CoreDNS SOA-front fixed certmagic zone-determination; the Pebble/Caddy image-tag + -strict + vv2.8.4 bugs were all found + fixed live.
3. **LE Phase 5 (renewal/rotation) — ROOT-DIAGNOSED, PENDING.** Pebble $\geq 2.8.0$ /set-renewal-info stores+serves a past ARI window (CONFIRMED), but certmagic v0.21.3's persistent 6h ARI cache blocks a fast deterministic renewal (/load
 - restart both read the cache). NEXT = the feasibility research docs/research/caddy_2110_ari_refetch_20260701/: does certmagic $\geq v0.25.1$ re-fetch ARI on reload? GO $\Rightarrow$ port the challtestsrv DNS provider to libdns v1.0.0
 - rebuild the Caddy image at $\geq 2.11.0$ + re-verify issuance+renewal. Detail in docs/research/pebble_set_renewal_info_20260701/ + letsencrypt_renewal_20260701/. Phase 4 (LE-staging token §11.4.10) + Phase 6 (prod cutover) OPERATOR-BLOCKED.
4. **P10 dynamic-VPN real-VPN-egress** remains operator-gated on gluetun WireGuard creds (§11.4.66); the fail-closed half is autonomously doable (boot dynamic stack, tunnel down $\rightarrow$ branded 503 + no leak) — booting reclaims :53128, single-owner (§11.4.119), free the running proxy first.
5. **Release (§11.4.40/§11.4.113/§11.4.151):** after LE Phase 5 (or an operator decision to ship issuance-only) + a full retest, cut `helix_proxy-0.1.0-dev-0.0.2` via GitHub AND GitLab CLIs (GitLab OPERATOR-BLOCKED — no GitLab remote) with the docs/releases/CHANGELOG_helix_proxy-0.1.0-dev-0.0.2.md (Rev 2), FF-only onto main.

6. Every change: TDD reproduce-first (§11.4.43/§11.4.115), all warranted test types (§11.4.169), paired §1.1 mutation, independent review → iterate-to-GO (§11.4.142/§11.4.125/§11.4.134), docs in sync (§11.4.60/§11.4.65/§11.4.106), operator resources untouched (§11.4.174: wg0-mullvad, lava-*, whoami:58080).